VIRAJ RAJESH PATIL

AI & SOFTWARE ENGINEERING | COMPUTER VISION | CYBERSECURITY

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PROFILE

Builder-focused B.Tech Electronics & Telecommunication student developing intelligent and secure digital products across computer vision, full-stack web, data engineering, and cybersecurity. Built a working YOLO/OpenCV surveillance prototype and a deployed mobile-first e-commerce platform; currently developing scalable AI data infrastructure and a hybrid personal AI companion.

SELECTED PROJECTS

AI Surveillance & Intrusion Monitoring System — **WORKING PROTOTYPE**

Python | OpenCV | Ultralytics YOLO | Flask | NumPy | Git

- Built a real-time webcam pipeline combining motion detection with confidence-scored person detection; added alert-hold and stability logic to reduce flicker and duplicate triggers.
- Developed a Flask dashboard that reports monitoring status and serves timestamped intrusion snapshots, validating the full capture-to-dashboard workflow locally.

BILLi&BoBA NAILS - Mobile-first E-commerce Platform — **DEPLOYED**

Next.js | TypeScript | Tailwind CSS | Supabase | Vercel

- Built a responsive storefront with product filtering, set details, cart and checkout journeys, photo-sizing guidance, order tracking, and customer/admin flows tailored for phone users.
- Implemented Supabase-backed admin media management and an auto-rotating photo/video preview, then refined glass UI, animated navigation, and production deployment on Vercel.

Interactive Developer Portfolio — **ACTIVE BUILD**

Next.js | TypeScript | Tailwind CSS | Framer Motion | React Three Fiber

- Creating a motion-led portfolio with a real-time 3D initial, cursor parallax, scroll transitions, a hanging identity-card interaction, and video-based portrait treatment.
- Structuring projects as visual case studies that explain the problem, implementation decisions, outcomes, and technical learning behind each build.

Pipeline Punch - Scalable AI Data Ingestion Pipeline — **IN DEVELOPMENT**

Python | Pandas | SQL | REST APIs | Data Validation | Logging

- Designing modular connectors for API and file sources with schema validation, record normalisation, structured error handling, and persistence for downstream AI workflows.
- Roadmap includes idempotent loads, retries, metadata and lineage, observability, automated tests, and reproducible local-to-cloud execution.

NOVA - Hybrid Personal AI Companion — **ARCHITECTURE PHASE**

Python | Local/Cloud LLMs | RAG | Vector Search | Privacy-first Memory

- Defining an offline/online assistant architecture for context-aware conversation, personal productivity, document retrieval, and user-controlled long-term memory.
- Planned progression: private local core, optional cloud reasoning, multimodal inputs, and an embedded desk companion with expressive on-device interactions.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Bash, SQL, HTML/CSS
AI & Data: OpenCV, YOLO, NumPy, Pandas, Matplotlib, scikit-learn fundamentals, data preprocessing, RAG/LLM systems
Web & Backend: Next.js, React, Tailwind CSS, Flask, Supabase, REST APIs, responsive UI development
Cybersecurity: Linux, networking fundamentals, reconnaissance, OWASP concepts, Burp Suite, Nmap, Amass, vulnerability reporting
Developer Tools: Git, GitHub, VS Code, Kali Linux, Ubuntu, VirtualBox, Vercel

PRACTICAL TRAINING & LABS

Cybersecurity Trainee | Tutedude - Project-based Online Training — **2026 | IN PROGRESS**

- Practising Linux administration, networking, Bash scripting, reconnaissance/OSINT, web application security, and privilege-escalation concepts in isolated lab environments.
- Using Kali/Ubuntu virtual machines, Git/GitHub, Burp Suite, Nmap, Amass, PortSwigger Academy, and TryHackMe to build repeatable technical workflows and concise lab notes.

EDUCATION & STRUCTURED LEARNING

B.Tech - Electronics & Telecommunication Engineering — **2023-2027**

Prof. Ram Meghe Institute of Technology & Research, Badnera-Amravati

HSC - Physics, Chemistry & Mathematics | Shri Shivaji Science College, Amravati — **2022-2023**

Ongoing learning: Tutedude Cybersecurity Training; PortSwigger Web Security Academy labs; TryHackMe cybersecurity labs; machine-learning mathematics and Python data stack